

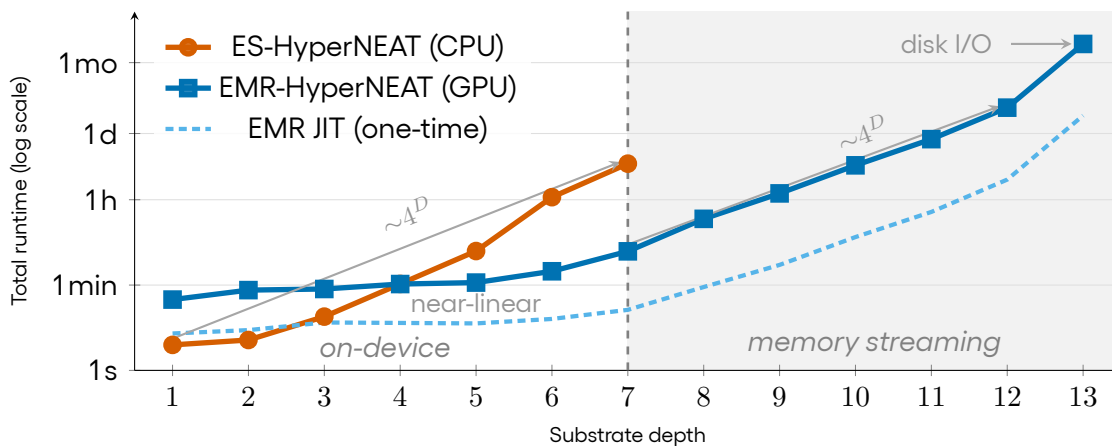
Evaluate everything, filter later.

Scaling adaptive substrate neuroevolution: sequential quadtree → parallel tensors

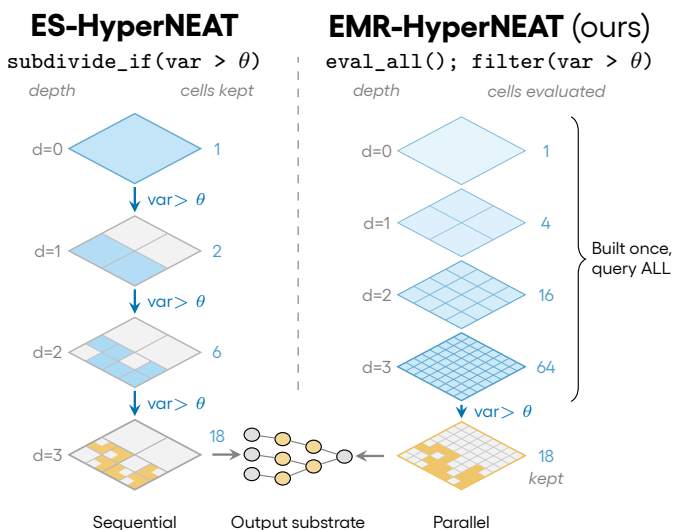
Tensor-Accelerated Eager Multi-Resolution Grids for Evolving Large-Scale Substrates

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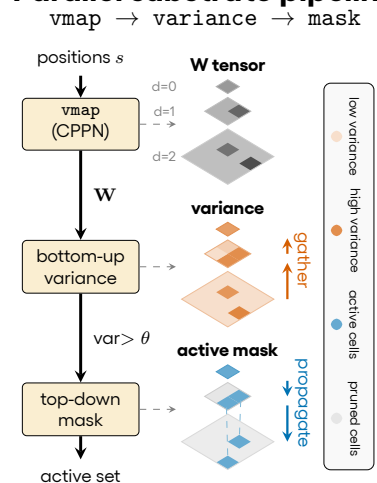
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Runtime over 30 generations on XOR; CPU: Apple M4 Max, GPU: NVIDIA RTX 2080 Ti 11GB.
ES-HyperNEAT grows as $\sim 4^D$, infeasible past depth 7; EMR-HyperNEAT stays near-linear on-device, then $\sim 4^D$ streaming from RAM at depths 7–12, much steeper on disk at depth 13.



Parallel substrate pipeline



The quadtree bottleneck

Children wait on parents, every CPPN grows a different tree, and dynamic shapes defeat batching and JIT.

What it unlocks

- One shared grid, static shapes ⇒ JAX JIT; reused across problems, batched over the population
- Substrates beyond GPU memory, streaming from RAM and disk
- Six substrate types, feedforward → full recurrent; the best type varies by problem

Guaranteed correct

Under identical thresholds, EMR discovers a *superset* of ES-HyperNEAT's positions; identical position sets produce identical networks.

It reaches high-variance regions that the quadtree prunes unseen, and it keeps every resolution level, not only leaves; in practice, equal or higher solve rates.

When it pays off

EMR trades redundant queries for parallelism: it wins on deep substrates, large populations, long runs; the quadtree, on small, short ones.

